4½ DIGIT DIGITAL MULTIMETER

Model 856

The Model 856 is an auto-ranging 4.5 digit digital multimeter. This portable DMM has all of the conventional voltage, current, resistance, continuity and

- *True rms AC Voltage and Current*
- *dBm Direct Reading and dB Relative*
- *Frequency Counter to 300 kHz*
- *Diode and Continuity Test Modes*
- *Auto-Ranging or Manual*
- *Three Level Comparator Function*
- *Data Hold, Key Lock and Last Memory*
- *Large Green LED and Bargraph Display*

diode measurement functions. Some of the additional standard features are comparison of measured results, frequency measurement and bargraph display and calculated functions for deviation, relative and decibel measurements. AC voltage and current measurements are made via a true rms measurement technique. The 3-level comparator allows measured results to be displayed as high, go or low with respect to preset upper and lower limits. AC voltage readings can be displayed in dBm or dB relative to an arbitrary level. The large green LED display is easy to read and gentle on the eyes.

1. Comparator function results (HI, GO, LO) as well as dBm, dB indicator.
2. 4.5 digit 7-segment LED readout.
3. AUTO RANGE and up-down, manual range selectors.
4. Bargraph display gives a visual indication of the range selected.
5. Recessed and protected input jacks.
6. Operation selectors: START/STOP for use with the comparator function; COMPARE sets up the comparison function; SELECT selects either the results or value of the deviation measurement function. It is also used to select dBm or dB in the dB function; GPIB (option) selects local or GPIB control; KEYLOCK locks the selected operation; RELATIVE sets up deviation measurements; HOLD holds the reading when pressed.
7. FUNCTION keys select DC A or V, AC A or V, Ω, diode test, audible continuity or FREQuency.

KEY SPECIFICATIONS

Measuring Functions
AC/DC voltage, AC/DC current, resistance, frequency, diode and continuity checks
Additional Functions
Deviation measurement, dBm measurement, dB measurement, comparator, range control, data hold, key lock, last memory and range display functions
Input Characteristics
Accuracy: ± % of reading ± digit

DC Voltage

Range	Full Scale	Resolution	Measuring Accuracy	Input Resistance
300mV	299.99mV	10µV	±0.03±2	>1000 MΩ
3V	2.9999V	100µV	±0.05±2	11 MΩ±5%
30V	29.999V	1mV	±0.05±2	10 MΩ±5%
300V	299.99V	10mV	±0.05±2	10 MΩ±5%
1000V	1000.0V	100mV	±0.05±2	10 MΩ±5%

AC Voltage (True rms)

Range	Full Scale	Resolution	Measuring Accuracy			Input Resistance
			40 Hz to 10 kHz	10 kHz to 20 kHz	20 kHz to 100 kHz	
3V	2.9999V	100µV	±0.4±30	±1±40	±5±100	11MΩ±5%
30V	29.999V	1mV	±0.4±30	±1±40	±5±100	10MΩ±5%
300V	299.99V	10mV	±0.4±30	±1±40	±5±100	10MΩ±5%
750V	750.0V	100mV	±0.1±30 (40Hz to 1kHz)			10MΩ±5%

DC Current

Range	Full Scale	Resolution	Measuring Accuracy	Input Resistance
300mA	299.99mA	10µA	±0.5±2	About 1.0Ω
3000 mA	2999.9mA	100µA	±0.7±2	About 0.1Ω

AC Current (True rms)

Range	Full Scale	Resolution	Measuring Accuracy (40 Hz to 1 kHz)	Input Resistance
300mA	299.99mA	10µA	±1.5±40	About 1.0Ω
3000 mA	2999.9mA	100µA	±1.7±40	About 0.1Ω

Resistance

Range	Full Scale	Resolution	Measuring Accuracy	Open Voltage
300Ω	299.99Ω	0.01Ω	±0.1±5	About 3.0V
3kΩ	2.9999kΩ	0.1Ω	±0.07±2	About 1.2V
30kΩ	29.999kΩ	1Ω	±0.07±2	About 1.2V
300kΩ	299.99kΩ	10Ω	±0.07±2	About 1.2V
3MΩ	2.9999MΩ	100Ω	±0.1±2	About 1.2V
30MΩ	29.999MΩ	1000Ω	±0.3±5	About 1.2V

Frequency

Range	Full Scale	Resolution	Measuring Accuracy
5Hz-1kHz	999.99Hz	0.01Hz	±0.02±2
10kHz	9.9999kHz	0.1Hz	±0.02±2
100kHz	99.999kHz	1Hz	±0.02±2
300kHz	300.00kHz	10Hz	±0.02±2

Diode Check

Range	Full Scale	Measuring Accuracy	Measuring Current	Open Voltage	Display Value
3V	2.9999V	±1.5±2	0.5 to 1 mA	About 3V	Forward Voltage

Continuity Check

Continuity Range	Reaction Speed	Continuity Check
0 to 100 ± 50 Ω	About 1 ms	Electronic Beeper

ENVIRONMENTAL

Operating Temperature
0°C to 40°C
23°C ± 5°C guaranteed accuracy
Humidity
Less than 80%

POWER REQUIREMENTS

100 to 240 V ac (manual switching)
50/60 Hz

PHYSICAL

Size (W x H x D)
7⅞ x 3 3/16 x 10¼ in.
200 x 80 x 260 mm
Weight
5.9 lbs., 2.7 kg

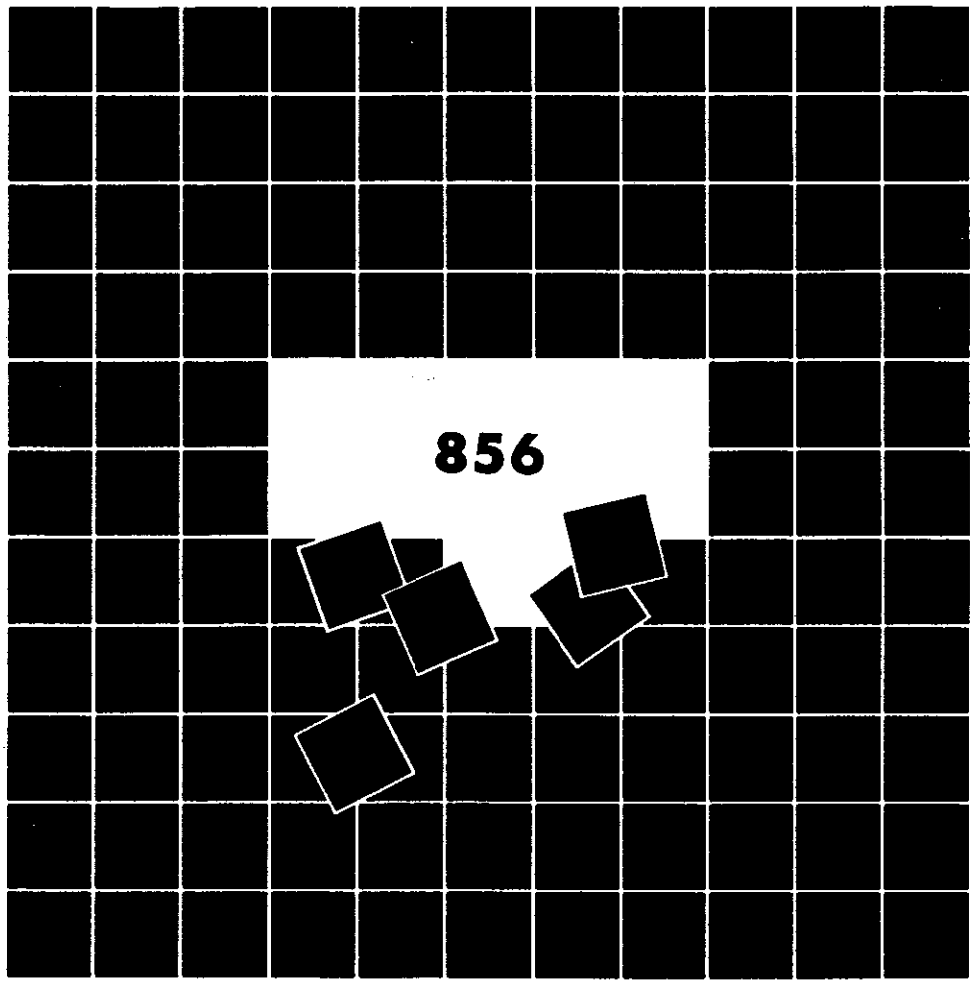
SUPPLIED ACCESSORIES

Test Leads
Fuse (250 V, 0.5 A, Time Lag)
Fuse (250 V, 3 A, Time Lag)

LEADER

MULTIMETER

INSTRUCTION MANUAL



LEADER ELECTRONICS CORP.

FOR SAFETY'S SAKE

Thank you for purchasing our product.
Please observe the following safety precautions when operating this instrument.

⚠ WARNING

- Do not remove any cases or covers.
  The high-voltage section inside this instrument can cause electrical shock.
- Do not operate this instrument and connected units in a volatile or flammable atmosphere.
  An explosive can result.
- Do not insert metal objects (e.g., wire, pin) into the vents.
  Otherwise, you may damage the instrument or suffer electrical shock.
- Connect this instrument to the rated power line voltage.
  Excessive voltage can cause fire.
- Do not touch the high-voltage section with hand directly when measuring it.
  You may suffer electrical shock.
- Do not connect this instrument to equipment whose chassis has electrical potential to ground (i.e., transformerless equipment).
  Otherwise, you may damage the instrument or suffer electrical shock.

⚠ CAUTION

- Use only the fuse of correct type and rating for replacement.
  Before replacing the fuse, be sure to turn the power switch off and disconnect the power cord from the mains.

Cautions on operation appear in the instruction manual. Read the manual carefully to ensure correct operation.

Explanation of the Terms

⚠ WARNING ... The WARNING calls attention to abnormal conditions or dangerous practices that could result in personal injury or death.

⚠ CAUTION .... The CAUTION calls attention to abnormal conditions or dangerous practices that could result in damage to the instrument or other property.

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1. SPECIFICATIONS

1.1 Introduction

The Model 856 is a 4.5-digit autoranging multimeter (maximum display 29999) which, in addition to basic voltage, current and resistance functions, has frequency measurement capability. Added functions include a comparision function which operates on measured results, and calculation functions (deviation measurements, dBm measurements, and dB measurements), making the Model 856 usable in a wide range of applications.

In addition, the Model 856 is housed in a metal case, thereby reducing unwanted radiation, and ensuring reliable measurements on receivers.

The Model 856 is a highly useful instrument not only in research and development applications but in a diverse range of applications including servicing and production-line measurements as well.

The Model 856 is available in four types, enabling selection for variation applications.

Model 856 ..... Basic model

Model 856G ... Basic model with the GPIB interface added

1.2 Features

Measurement Features

- Basic accuracy of 0.03% for DC voltage and 0.4% for AC voltage
- Both autoranging and manual range settings are possible.
- AC voltage and current measurements are true RMS measurements, enabling accurate RMS measurements on even distorted waveforms.
- AC voltage can be read directly in dBm units (600 Ω), a function convenient in level measurements in audio systems.
- Decimal measurements of AC voltage with reference to an arbitrary input voltage as 0 dB are possible. This function is convenient for use in generating frequency response graphs and in making amplifier gain measurements (AC measurement accuracy of 20 kHz: ±1% ±40 digits or greater).
- Frequency measurements are made with high speed using the reciprocal technique. The maximum resolution is 0.01 Hz and the measurement range is 5 Hz to 300kHz.
- Deviation measurements are possible with respect to any arbitrary value, enabling high-accuracy resistance measurements by compensating for lead resistance. This also enables voltage and current deviation measurements.
- A comparator function is provided for convenient use in parts selection, and in voltage and current checks on the production line.

Operating Features

- A large green LED display provides easy-to-read measurement values.
- A data hold function simplifies the data-taking task.
- A keylock function prevents misoperation during measurements.
- A last memory function eliminates the need to make measurement settings on the production line by saving the measurements when power is switched off.
- An LED bar display indicates the measurement range, enabling verification that the proper range has been selected. For frequency measurements, this serves as an input-signal attenuation monitor.
- The line voltage can be selected in the range 100 to 240 V by a rear-panel switch, making the Model 856 usable virtually anywhere in the world.

1.3 Performance Specifications

(1) Measurement Functions — DC voltage, AC voltage, DC current, AC current, resistance, diode check, continuity check, frequency

(2) Input Characteristics — Measurement accuracy indicated as (±% of reading ± digits).

DC Voltage

Range	Full Scale	Resolution	Measurement Accuracy	Input Resistance
300 mV	299.99 mV	10 µV	±0.03 ±2	1000 MΩ min.
3 V	2.9999 V	100 µV	±0.05 ±2	11 MΩ ±5%
30 V	29.999 V	1 mV	±0.05 ±2	10 MΩ ±5%
300 V	299.99 V	10 mV	±0.05 ±2	10 MΩ ±5%
1000 V	1000.0 V	100 mV	±0.05 ±2	10 MΩ ±5%

AC Voltage (True RMS Value)

Crest factor:3

Range	Full Scale	Resolution	Measurement Accuracy			Input Resistance
			40 Hz-10kHz	10 kHz-20 kHz	20 kHz-100 kHz	
3 V	2.9999 V	100 µV	±0.4 ±30	±1 ±40	±5 ±100	11 MΩ ±5%
30 V	29.999 V	1 mV	±0.4 ±30	±1 ±40	±5 ±100	10 MΩ ±5%
300 V	299.99 V	10 mV	±0.4 ± 30	±1 ±40	±5 ±100	10 MΩ ±5%
750 V	750.0 V	100 mV	±1.0	±30 (40 Hz to 1kHz)		10 MΩ ±5%

Measurement Accuracy: More than 10% of full scale

DC Current

Range	Full Scale	Resolution	Measurement Accuracy	Input Resistance
300 mA	299.99 mA	10 µA	±0.5 ±2	Approx. 1.0 Ω
3000 mA	2999.9 mA	100 µA	±0.7 ±2	Approx. 0.1 Ω

AC Current (True RMS Value)

Crest factor:3

Range	Full Scale	Resolution	Measurement Accuracy 40 Hz to 1 kHz	Internal Resistance
300 mA	299.99 mA	10 µA	±1.5 ±40	Approx. 1.0 Ω
3000 mA	2999.9 mA	100 µA	±1.7 ±40	Approx. 0.1 Ω

Measurement Accuracy: More than 10% of full scale

Resistance

Range	Full Scale	Resolution	Measurement Accuracy	Open-Circuit Voltage
300 Ω	299.99 Ω	0.01 Ω	±0.1 ±5	Approx. 3.0 V
3 kΩ	2.9999 kΩ	0.1 Ω	±0.07 ±2	Approx. 1.2 V
30 kΩ	29.999 kΩ	1 Ω	±0.07 ±2	Approx. 1.2 V
300 kΩ	299.99 kΩ	10 Ω	±0.07 ±2	Approx. 1.2 V
3 MΩ	2.9999 MΩ	100 Ω	±0.1 ±2	Approx. 1.2 V
30 MΩ	29.999 MΩ	1000 Ω	±0.3 ±5	Approx. 1.2 V

Frequency

Range	Full Scale	Resolution	Measurement Accuracy
5Hz to 1kHz	999.99 Hz	0.01 Hz	±0.02 ±2
1kHz to 10kHz	9.9999 kHz	0.1 Hz	±0.02 ±2
10kHz to 100kHz	99.999 kHz	1 Hz	±0.02 ±2
100kHz to 300kHz	300.00kHz	10 Hz	±0.02 ±2

Diode Check

Range	Full Scale	Measurement Accuracy	Measurement Current	Open-Circuit Voltage	Measurement Display
3 V	2.9999 V	±1.5 ±2	0.5 to 1 mA	Approx. 3 V	Forward voltage

Continuity Check

Resistance Range	Response Speed	Continuity Indication
0 to 100 ±50 Ω	Approx. 1 ms	Electronic beeper

(3) Sampling Rate — 2 to 3 times/s

(4) Added functions — Deviation measurement, dBm measurement, dB measurement, range control, data hold, keylock, backup, and range display functions.

Calculation equations

Deviation measurement — (A/D conversion result) − (compensation value) = (display)

dBm measurement

$$20 \log \frac{(A/D\ conversion\ results)}{0.7746\ V\ (1\ mW\ into\ 600\ \Omega)} = (Display)$$

dB measurement

$$20 \log \frac{(A/D\ conversion\ results)}{(Arbitrary\ AC\ voltage\ input)} = (Display)$$

(5) Comparator function

Comparison range	"LO" < lower limit ≦ "GO" ≦ upper limit < "HI"

(6) Display

Type	Green LED
Maximum displayed value	29999 (4.5 digits) (99999 for frequency measurements)
Input overflow indicator	"OF" display
Polarity	Automatic switching ("−" displayed for negative values)

(7) Range switching Automatic/manual

(8) Maximum allowable applied voltage/current ⚠ CAUTION

Measurement Function	Maximum Allowable Applied Voltage/Current
DC voltage AC voltage Frequency	1000 VDC or 750 VAC
DC current AC current	3A, 250 VAC
Resistance Diode check Continuity check	250 VAC

(9) Electronic beeper

A beeper sounds for key input, continuity check, and the range error condition.

(10) Environmental conditions

0 to 40° C (23 °C ± 5° C for guaranteed performance)

80% max. humidity

(11) Power requirements

100 to 240 VAC (manually switched), 50/60 Hz

(12) General specifications

Environmental conditions	
Operating Temperature	**: 0～40℃**
Humidity	**: 80%RH (without Condensation)**
Spec-Guaranteed Temperature	**: 18～28℃**
Humidity	**: 80%RH (without Condensation)**
Operating Environment	**: Indoor use**
Operating Altitude	**: up to 2,000m**
Overvoltage Category	**: Ⅱ**
Pollution degree	**: 2**
Dimensions	**200(W)×80(H)×260(D) mm**
Weight	**2.7 kg**
Accessories	**Test leads (black, red) ····· 1 set**
	Power line fuse (250 V / 0.5A midget time-lag type)····· 1
	Power current fuse (250 V / 3A midget time-lag type with arch suppressant) ····· 1
	Power cord ····· 1
	Instruction manual····· 1

2. PANEL DESCRIPTIONS

In this manual, circled numbers refer to knobs and terminals on the front panel shown in the diagrams below.

2.1 Front Panel

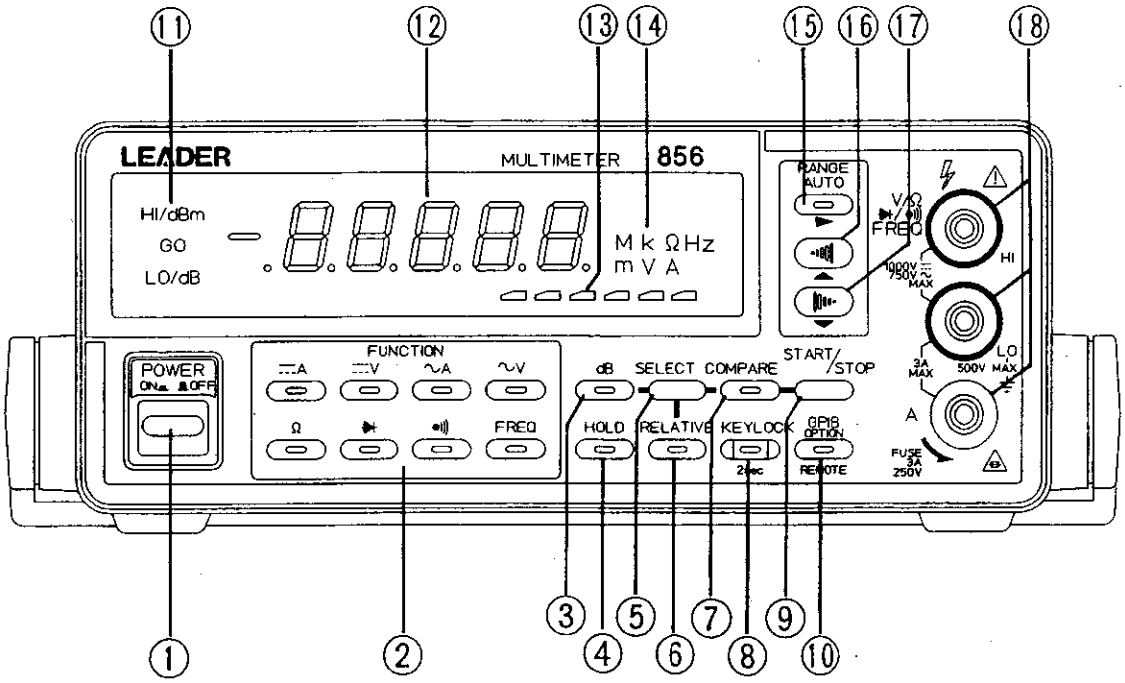


Fig. 2-1

① POWER switch (push button switch)

② FUNCTION selection switches

These switches are used to select the following functions.

- ⎓ V DC voltage
- ∼ V AC voltage
- ⎓ A DC current
- ∼ A AC current
- Ω Resistance
- FREQ Frequency
- ⊣⊢ Diode check
- ·))) Continuity check

When any switch is pressed, the corresponding function is selected, and the corresponding function indicator lights.

③ dB switch (key switch)

This is the dBm/dB function switch.

When the ~ V function is selected, if this switch is pressed, the dBm/dB function operates.

④ HOLD switch (key switch)

This is the data hold function switch.

When it is pressed, the data hold function operates.

⑤ SELECT switch (key switch)

This switch selects either the deviation results or deviation value display for the deviation measurement function. It is used to switch between dBm and dB for the dBm/dB function.

In the comparator function setting mode, it is used to select between HI and LO.

⑥ RELATIVE switch (key switch)

This is the deviation measurement function switch.

When the switch is pressed, the deviation measurement function operates.

⑦ COMPARE switch (key switch)

This is the comparator function switch.

When the switch is pressed, the comparator function setting mode is entered.

⑧ KEY LOCK switch (key switch)

This is the key lock function switch.

When the switch is pressed, the key lock function operates. When this function is operative, the indicator lights.

⑨ START/STOP switch (key switch)

This is the start/stop switch for the comparator function.

⑩ LOCAL switch

This switch is used to switch from the GPIB interface remote control mode to the local mode.

When the remote control mode has been enabled. the indicator lights.

⑪ HI/dBm, GO, LO/dB indicator

This indicates the type of comparator function results.

It indicates the units for the dBm/dB function as well.

⑫ Measurement-value display (7-segment LED)

This display indicates measured values, overflow messages, and error messages.

⑬ Range indicator (bar LED display)

This enables a check of the measurement range. In addition, for frequency measurements, it serves as an input attenuation monitor.

⑭ Units indicator

This indicators the measurement units.

⑮ RANGE AUTO switch (key switch)

In the ⎓V, ∼V, Ω, ⎓A, and ∼A functions, this enables switching from autoranging to manual range selection.

In the autoranging mode, the indicator lights.

⑯ switch (key switch)

In the ⎓V, ∼V, Ω, ⎓A, and ∼A functions, this sets the range for manual range selection.

Each time the switch is pressed, the range moves up.

For the FREQ function, this switches the attenuator.

⑰ switch (key switch)

In the ⎓V, ∼V, Ω, ⎓A, and ∼A functions, this sets the range for manual range selection.

Each time the switch is pressed, the range moves down.

For the FREQ function, this switches the attenuator.

⑱ Input terminals

These are the signal input terminals. The current terminal is provided with a built-in protective fuse.

2.2 Rear Panel

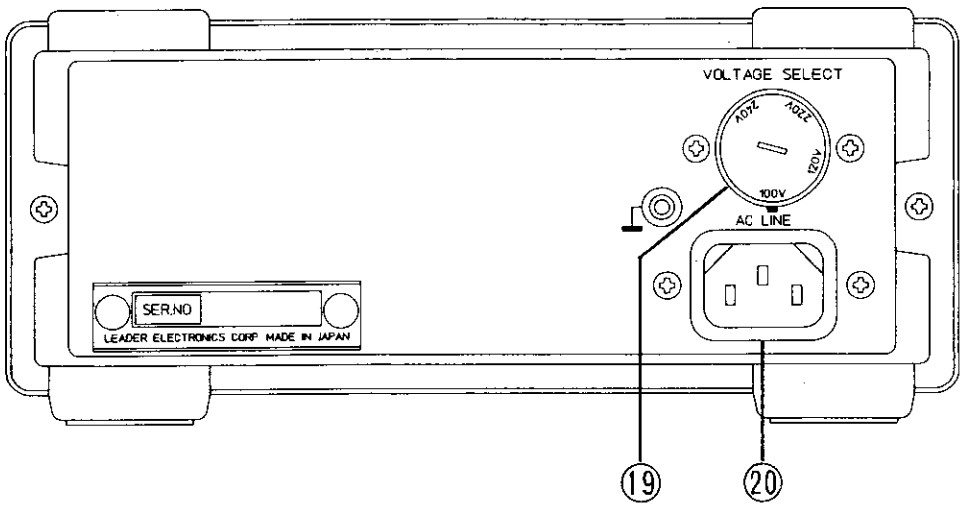


Fig. 2-2

⑲ **VOLTAGE SELECT**

This is the line voltage selector. Set it to the appropriate line voltage.

⑳ **AC power line connector**

The AC power cord is connected at this connector.

3 OPERATION

⚠ CAUTION

3.1 Operating Precautions

(1) Be sure to set the line voltage properly.

Set the VOLTAGE SELECT switch on the rear panel to the line voltage to be used.

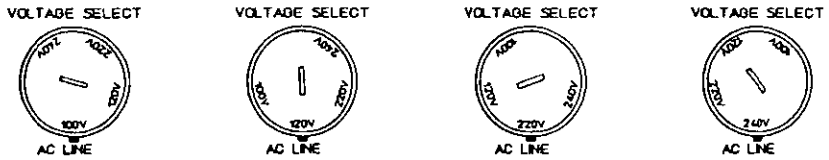


Fig. 3-1

(2) Do not apply excessive inputs to the input terminals which would exceed the maximum withstanding current and voltage ratings of this instrument.

This type of excessive input risks damage to input circuitry and components.

For information on maximum voltage and current ratings, refer to Section 1.3 (8), Maximum allowable applied voltage/current.

(3) Do not used this instrument in locations which are subject to high temperature and humidity.

Operate the instrument in a temperature range of 0° C to +40° C, and in a humidity range of 10 to 80%. Use under severe environmental conditions can cause failures.

(4) Do not subject the instrument to shock.

When carrying this instrument, take care not to drop it, and that it does not strike other objects or is otherwise subjected to excessive shock, as this type of treatment can cause failures and misoperation.

(5) Do not pull on the power cable or test leads.

When removing the power cable or test leads, hold them by the connector. If you forcibly pull the cable or test leads, opens or bad connections can occur, resulting in failures.

(6) Panel last memory

Panel settings are held in memory even when the power is switched off by virtue of a panel last memory function.

The panel last memory function operates for approximately one month after operating the instrument for eight hours. However, care is required as the backed up memory contents will be cleared if power is switched off immediately after power is applied and initialization is done (i.e. when all the display lamps are extinguished).

(7) Fuse replacement

If a fuse should blow, after removing the cause of the blown fuse and making a safety check, replace the fuse, following the instructions in Section 7 (Fuse Replacement). Always use the specified type of fuse, and under no circumstances attempt to replace the fuse by wire.

3.2 Panel Markings

The Model 856 panel has the markings described below.

Terminal markings

⚡ ... There is a danger of high-voltage at the input terminals. Take sufficient precaution.

⚠ ... Depending upon use, internal circuitry may be damaged. Before using, be sure to read Section 3.1 (Operating Precautions)

⚠ ... Internally fused. If the fuse should blow due to excessive current, replace, referring to Section 7 (Fuse Replacement).

⏚ ... Ground

Function markings

~ ... AC signal

⎓ ... DC signal

⊳| ... Diode test

•)) ... Continuity test using a beeper

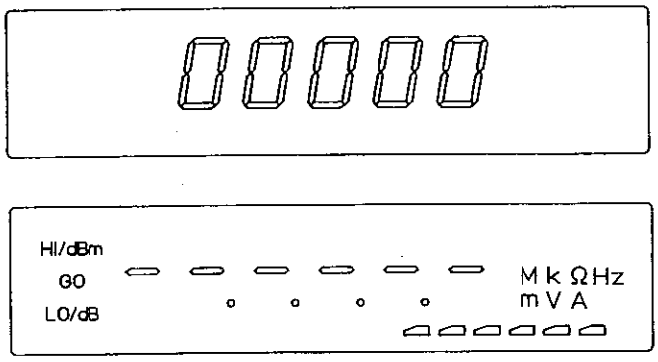
3.3 Measurement

3.3.1 Measurement Procedure

The general measurement procedure is as follows.

(1) Set the POWER switch ① to ON.

The displays will begin to light, and then last memory settings will be made.



(2) Using the FUNCTION switches ②, select the measurement function.

- DC voltage ⎓ V
- AC voltage ∼ V
- DC current ⎓ A
- AC current ∼ A
- Resistance Ω
- Frequency FREQ
- Diode check ⊣⊢
- Continuity check ·))

In making the function selection, the range switching is autorange. At this point in the procedure, the range switching is autorange.

To use manual range selection, press the RANGE AUTO switch ⑮ (indicator extinguishes).

(3) Connect the test leads to the input terminals ⑱ according to the measurement function selected.

Table 3-1

Function	Red Lead	Black Lead
Voltage Resistance Frequency Diode check Continuity	HI terminal	LO terminal
Current	A terminal	LO terminal

Precaution in measurement AC voltage and current

In particular when measuring low-level AC voltages and currents, the measurement leads may take up external noise. To avoid this, twist the red and black leads when making such measurements.

(4) Connect the test leads to the measurement points.

The maximum inputs to the input terminals are as follows. Take care that these limits are not exceeded.

Between HI and LO terminals	1000 VDC, 750 VAC
Between A (current) and LO terminals	3 A (protective fuse provided), 250 V

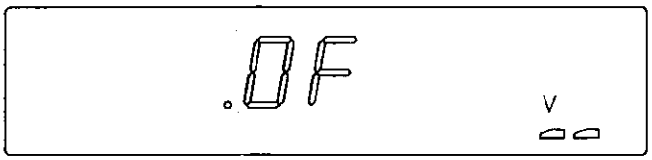
(5) The measured value will be indicated on the measurement value display ⑫.

The units of the display will be indicated by the units indicator ⑭.

Note that for positive values of DC voltage and current, the polarity (+) is not displayed. The minus sign (−) will be displayed, however, for negative values.

In making manual-range measurements, if the measured value display ⑫ indicates "OF", an input overrange has occurred. If this occurs, press the switch ⑯ to increase the range. However, in making resistance measurements, the following conditions must also be considered.

- Test leads not making contact with the measurement points
- Open condition in the circuit under measurement



When the display indicates " 2999" or lower, press the switch ⑰ to decrease the range and increase the displayed value. Because this instrument is a 4.5-digit instrument (with a maximum display of 29999), you should try to make full use of the display capacity.

3.3.2 Function Switching and Full Scale

DC Voltage Measurement

When the ⎓V of the FUNCTION switches ② is pressed, the DC voltage measurement mode is selected.

The relationship of measurement range to full-scale value and bar LED display is shown in Table 3-2.

Table 3-2

Range	Full-Scale Value	Bar LEDs
300 mV	299.99 mV	
3 V	2.9999 V	
30 V	29.999 V	
300 V	299.99 V	
1000 V	1000.0 V	

AC Voltage Measurement

When the ∼V of the FUNCTION switches ② is pressed, the AC voltage measurement mode is selected.

The relationship of measurement range to full-scale value and bar LED display is shown in Table 3-3.

Table 3-3

Range	Full-Scale Value	Bar LEDs
3 V	2.9999 V	
30 V	29.999 V	
300 V	299.99 V	
750 V	750.0 V	

DC Current Measurement

When the ⎓A of the FUNCTION switches ② is pressed, the DC current measurement mode is selected.

The relationship of measurement range to full-scale value and bar LED display is shown in Table 3-4.

Table 3-4

Range	Full-Scale Value	Bar LEDs
300 mA	299.99 mA	
3000 mA	2999.9 mA	

AC Current Measurement

When the ~ A of the FUNCTION switches ② is pressed, the AC current measurement mode is selected.

The relationship of measurement range to full-scale value and bar LED display is shown in Table 3-5.

Table 3-5

Range	Full-Scale Value	Bar LEDs
300 mA	299.99 mA	
3000 mA	2999.9 mA	

Resistance Measurement

When the Ω of the FUNCTION switches ② is pressed, the resistance measurement mode is selected.

The relationship of measurement range to full-scale value and bar LED display is shown in Table 3-6.

Table 3-6

Range	Full-Scale Value	Bar LEDs
300 Ω	299.99 Ω	
3 kΩ	2.9999 kΩ	
30 kΩ	29.999 kΩ	
300 kΩ	299.99 kΩ	
3 MΩ	2.9999 MΩ	
30 MΩ	29.999 MΩ	

Frequency Measurement

When the FREQ of the FUNCTION switches ② is pressed, the frequency measurement mode is selected.

The range in this mode is automatically selected using autoranging, and the input sensitivity range is manually selected.

By using the switch ⑯ and ⑰ to adjust the sensitivity according to the input voltage, the frequency will be displayed.

Each time the switch ⑯ is pressed, the attenuation ratio increases by ten-fold with setting possible to 1/1000.

Each time the switch ⑰ is pressed, the attenuation ratio decreases by ten-fold with setting possible to 1/1.

The optimum ranges for input levels are shown in Table 3-7.

Table 3-7

Attenuator	Maximum Sensitivity	Maximum Input Sensitivity	Bar LEDs
1/1	225 mV RMS	±3 V Peak	
1/10	2.25 V RMS	±30 V Peak	
1/100	22.5 V RMS	±300 V Peak	
1/1000	225 V RMS	±1000 V Peak	
1/10000	Not specified	±1000 V Peak	

Diode Measurements

When the ⊣⊢ switch of the FUNCTION switch ② is pressed, the diode measurement mode is selected. In this mode measurement of the forward voltage drop across a diode is possible.

Continuity Check

When the ·)) switch of the FUNCTION switch ② is pressed, the continuity check mode is selected. In this mode,a beeper sounds when the resistance between two measured points is lower than approximately 50 Ω.

3.3.3 Auto/Manual Range Switching

This instrument enables both manual and automatic range switching.

Autoranging Measurements

After selecting ⎓ V, ∼ V, ⎓ A, ∼ A, or Ω function, the autoranging mode is selected automatically. When the LED of the RANGE AUTO switch ⑮ is extinguished, the RANGE AUTO switch can be pressed to select autoranging.

Manual Range Measurements

Press the RANGE AUTO switch ⑮ (to extinguish the LED) and then press the ⊣| switch ⑯ or |⊢ switch ⑰ to select the desired range.

Each time the ⊣| switch ⑯ is pressed, the range is increased, and each time the |⊢ switch ⑰ is pressed, the range is decreased, with setting being possible to the maximum or minimum range provided.

4. ADDED FUNCTIONS

4.1 dBm/dB Measurement

dBm Function

Using the dBm function, the level in a transmission system terminated by 600 Ω is displayed in dBm units. The dBm function can be used in such applications as level measurements on audio equipment.

With the ~ V function selected, when the dB switch ③ is pressed, "HI/dBm" above the display lights, and the dBm function operates. The dBm function displays results calculated as follows.

$$20\ log\ \frac{(A/D\ conversion\ results)}{0.7746V\ (600\Omega,\ 1mW)} = (Displayed\ value)$$

dB Function

With the dBm function selected, if the SELECT switch ⑤ is pressed, the "LO/dB" above the display lights and the function is changed to the dB function.

In the dB function, the input voltage at the time the SELECT switch ⑤ is pressed is taken as 0 dB, and the amount of variation from this value is displayed in units of dB. The dB function can be used in such applications as the generation of frequency response graphs and measurements of amplifier gain.

The dB function displays results calculated as follows.

$$20\ log\ \frac{(A/D\ conversion\ results)}{(When\ the\ SELECT\ switch\ ⑤\ is\ pressed)} = (Displayed\ value)$$

Thereafter, each time the SELECT switch ⑤ is pressed, alternate switching is performed between the dBm and dB functions.

Note: If an A/D conversion overflow occurs when the dBm/dB function is selected, "Error" is displayed, indicating a calculation result error.

dBm/dB Function Cancellation

The dBm/dB function can be canceled by any of the following methods.

- By pressing the dB switch ③ once again.
- By pressing the FUNCTION switch ②.
- By pressing the RELATIVE switch ⑥.
- By pressing the COMPARE switch ⑦.

4.2 Deviation Measurement (RELATIVE)

Relative data measurements are possible for all functions, with the exception of the frequency (FREQ) and continuity measurement (·))).

The RELATIVE measurement function measures the deviation from a reference which is an arbitrary measured data value. This function can be used in making high-accuracy resistance measurements by canceling out the lead resistance, and in measurements of voltage and current variations.

When the RELATIVE switch ⑥ is pressed, the deviation measurement function is selected.

The data at the time the RELATIVE SWITCH ⑥ is pressed is stored, the following calculation is performed, and the results are displayed.

(Measured value) − (Stored data value) = (Displayed value)

The stored data value can be in the range 0 to ±29999 counts.

The deviation result is displayed in the range ±59998 counts.

During operation of this function, an indicator lights, the measurement range is held, and changes in measurement range are not possible. In this mode, if the SELECT switch ⑤ is pressed, the indicator is extinguished, and the stored data value is displayed.

If the SELECT switch ⑤ is pressed once again, a change is made to the deviation result display.

Note: Should a measured value cause an overflow during a deviation measurement mode, "Error" is displayed to indicate a deviation result error.

If a measured value causes an overflow, pressing the RELATIVE switch ⑥ will not enable the deviation measurement function.

Deviation Measurement Mode Cancellation

The deviation measurement mode can be canceled by any of the following methods.

- By pressing the RELATIVE switch ⑥ once again.
- By pressing the FUNCTION switch ②.
- By pressing the dB switch ③ (when the ∼V function is operating).
- By pressing the COMPARE switch ⑦.

4.3 Hold Function

The data hold function operates for all functions with the exception of the continuity measurement function (·))).

Using the data hold function, it is possible to hold a displayed value for as long as necessary.

When the HOLD switch ④ is pressed, the data hold function operates.

During operation of this function, an indicator lights, and displayed data is held.

In this mode, all switch input other than the HOLD switch ④ is disabled.

Data Hold Function Cancellation

To cancel the data hold function, press the HOLD switch ④ once again.

4.4 Keylock Function

The keylock function disables all switch operation except for certain switches, thereby preventing misoperation during measurement.

When the KEYLOCK switch ⑧ is pressed, the keylock function operates.

During operation, an indicator lights. In this mode, all switch input other than HOLD switch ④ is disabled.

Keylock Function Cancellation

The keylock function can be canceled by depressing the KEYLOCK switch ⑧ for approximately two seconds.

5. COMPARATOR FUNCTION

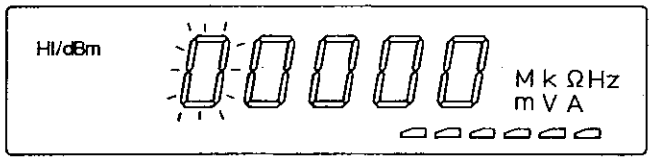
The comparator function operates for all measurement functions other than frequency and continuity.

This function checks whether the measured value is within a preset range. The comparator function can be used in such applications as parts selection, and in voltage and current checking on the production line.

Upper-Limit Setting

When the COMPARE switch ⑦ is pressed, the comparator function upper limit setting mode is entered. The comparator function indicator flashes, and the HI/dB indicator lights.

In this condition, the upper most order digit of the display value flashes.

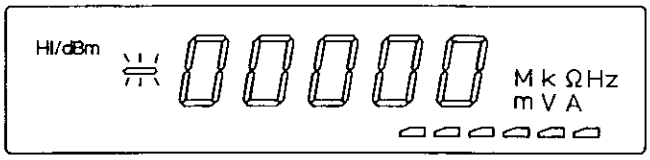


Use the ⑯ or ⑰ to set the uppermost order digit.

Next, press the RANGE AUTO ⑮ to move the flashing digit position down by one digit and make the same type of setting. Repeat this setting procedure until the desired setting value is input completely.

When the lowermost order digit is flashing, if the RANGE AUTO switch ⑮ is pressed, the lowermost order digit setting is completed, and polarity setting is possible. Press the switch ⑰ to select "−", or the switch ⑯ to select the "+". However, note that "+" will not be displayed.

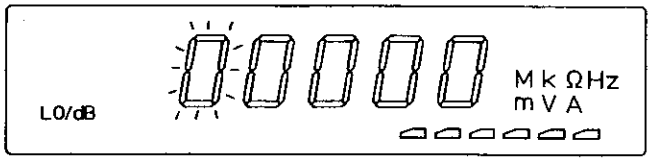
To perform a numerical setting, press the RANGE AUTO switch ⑮.



Lower-Limit Setting

After making the upper-limit value setting, if the SELECT switch ⑤ is pressed, the HI/dBm indicator is extinguished, the LO/dB indicator lights, and the lower-limit value setting mode is enabled.

As was done above for the upper limit value, use the RANGE AUTO switch ⑮ and the switch ⑯ switch ⑰ to set the input value.



Starting the Comparator Function

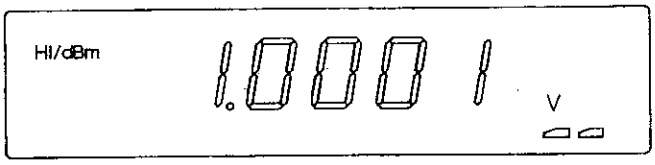
After making settings, when the START/STOP switch ⑨ is pressed, the comparator function starts.

The comparator function indicator will change from the flashing display to a constantly lighted display.

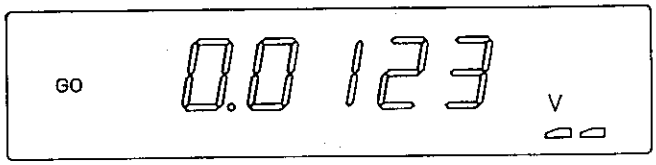
When the measured value is within the set range, the GO indicator will light.

(Example) Upper limit: +1.0000 V lower limit: −1.0000 V settings

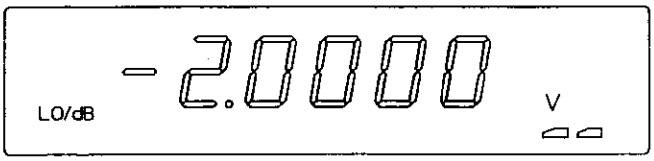
- Input: +1.0001 V



- Input: +0.0123 V



- Input: −2.0000V



Stopping the Comparator Function

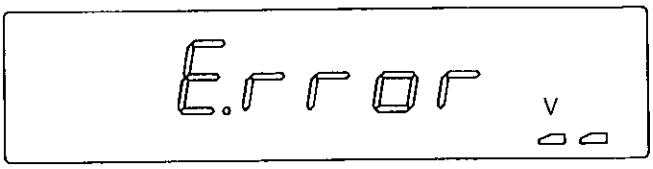
If the START/STOP switch ⑨ is pressed once again, the comparator function is stopped.

The comparator function indicator changes to a flashing display.

In this mode, if the RANGE AUTO switch ⑮ is pressed, the setting mode is entered once again.

Note: After setting the upper and lower limit values, if the START/STOP switch ⑨ is not pressed, the set values will not be stored.

When starting the comparator function, if the LO setting is higher than the HI setting, an A/D conversion result overflow will result, as indicated by a "Error" display.



Comparator Function Cancellation

The comparator function can be canceled by any of the following methods.

- By pressing the COMPARE switch ⑧.
- By pressing the FUNCTION switch ②.
- By pressing the RELATIVE switch ⑤.
- By pressing the dB switch ④ (when in the ~V function).

6. HANDLE

The angle of the handle can be set for various applications.

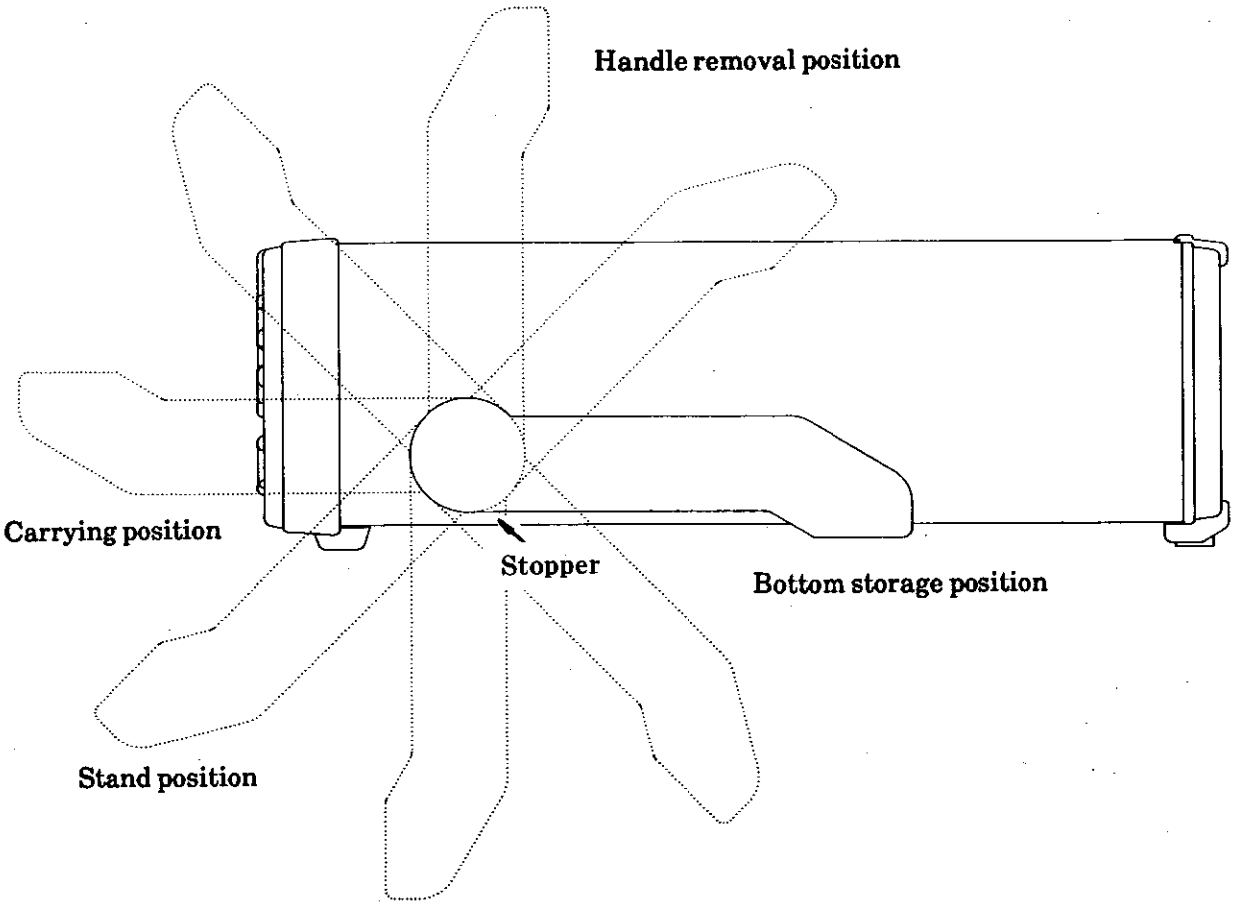


Fig. 6-1

The handle can be used as a stand as well.

The lock on the handle can be released by pulling the stoppers at the left and right outward using both hands, in which condition the handle can be rotated freely. When released, the handle locks once again.

When mounting this instrument in a rack, the handle is removed.

To remove the handle, rotate the handle to the "handle removal position" in Fig. 6-1, and pull the left and right stoppers out using both hands. In this condition, the handle can be removed from the mainframe by pulling towards the front panel side.

7. FUSE REPLACEMENT

Fuse Ratings

Never use fuses of other than the specified ratings.

Line fuse	250 V/0.5 A midget time-lag fuse
Current fuse	250 V/3 A midget fuse with arc suppression

7.1 Front Panel Fuse

Current Measurements

When the measured current exceeds the rated limit, the protective fuse on the front panel will blow. Since this prevents further measurements, the fuse must be replaced with the accessory spare fuse provided.

Replacement of the Protective Fuse

(1) Rotate the current terminal counterclockwise using the palm of the hand or a coin.

(2) Remove the terminal and the fuse held within the terminal.

(3) After replacing the fuse, insert the terminal once again and rotate clockwise.

(4) When the terminal is locked into place, measurements are once again possible.

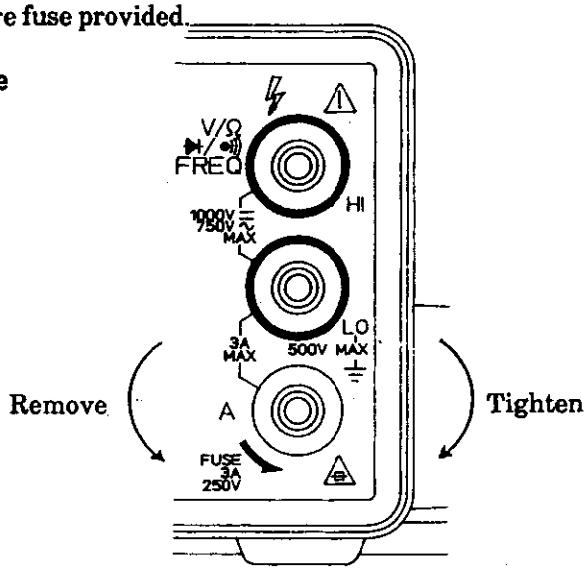


Fig. 7-1

8. MAINTENANCE

The Model 856 is designed to provide stable performance when used properly. If the instrument requires adjustment or calibration after extended use, be sure to contact your nearest Leader agent.

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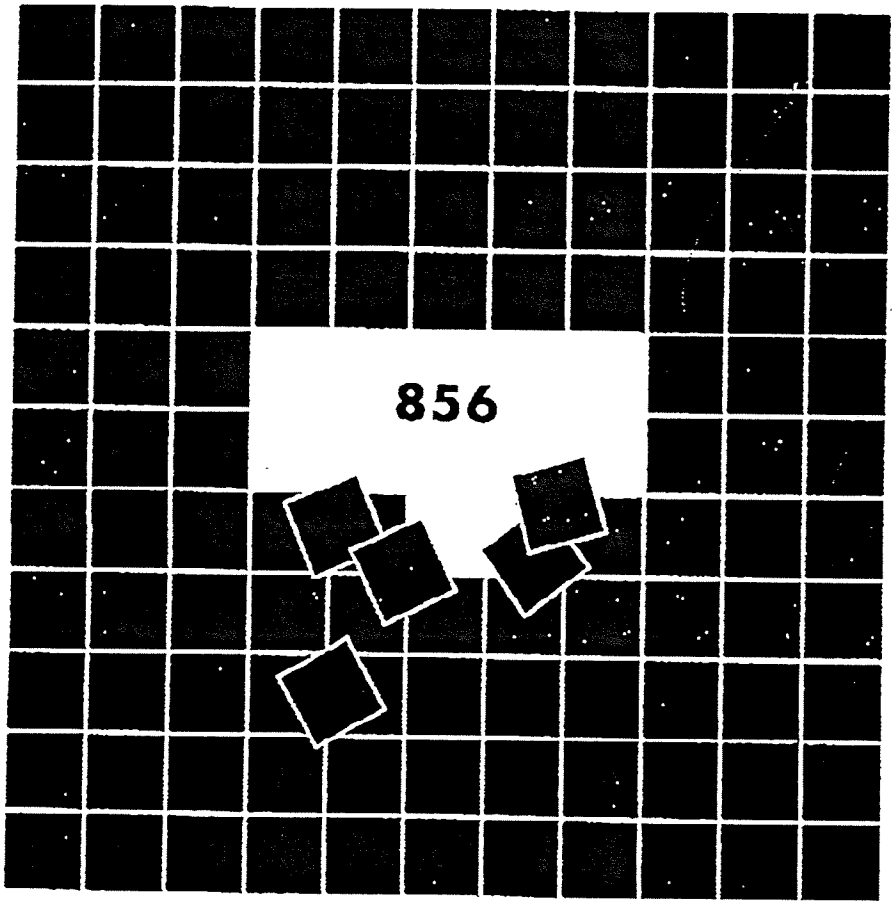
MULTIMETER

SERVICE MANUAL



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ADJUSTMENT/REPAIR MANUAL FOR MODEL 856

1. POWER LINE & FUSE

1) POWER LINE:
AC 100V, 120V, 220V & 240V can be selected with switch on rear panel. Make it sure that power line switch is OFF when power line cable is connected.

2) FUSE:
Two kinds of fuse are used:

2-1) 3A 250V ø5.2 x 20mm Fuse in glass tube of arc-extinguishing material is used to protect DMM circuit against overload input in current function. Fuse should be exchanged under test lead removed from input terminal and power switch to be OFF.

2-2) 0.5A 250V ø5.2 x 20mm Time lag Fuse in glass tube is used to avoid overload current. Fuse should be exchanged under power line cable removed from plug socket.

2. BACK-UP FUNCTION

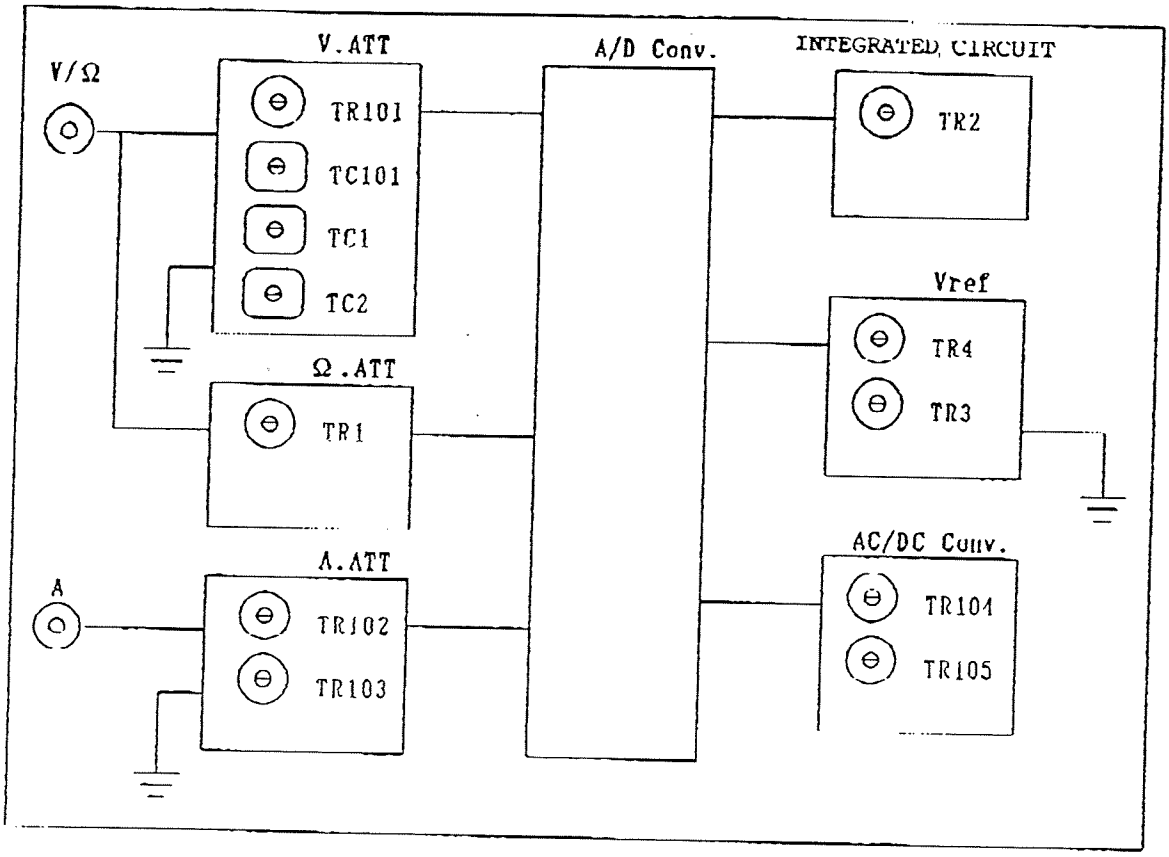
Panel setting is memorised by back-up function despite power OFF. After non-use for long period, back-up fuention must be checked under due aging. Back-up period is about one month after 8-hour operation.

Note: When power is OFF under all lighted indication, operation become initial stage without back-up function. Then DCV function starts under auto-range when power is ON.

3. ADJUSTMENT

Adjustment should be made under environmental conditions below 23°C ± 1°C 70% R.H. and under aging of 30 minutes or more after power ON.

1) ADJUSTMENT OF TRIMMER POSITION



2) ADJUSTMENT CONTENTS

TR101 : To adjust attenuator ratio for voltage attenuator (all ranges at DCV, ACV are influenced.)

TC101 : To adjust correcting capacitor of frequency response parallelled to input resistor 10MΩ for voltage attenuator (all ranges of frequency response at ACV are influenced.)

TC1 : To adjust correcting capacitor of frequency response parallelled to 3V range resistor 1.1MΩ for voltage attenuator (3V range of frequency response at ACV is influenced.)

TC2 : To adjust correcting capacitor of frequency response parallelled to 30V range resistor 101KΩ for voltage attenuator (30V range of frequency response at ACV is influenced.)

TR1 : To adjust reference resistor of 3000KΩ , 30MΩ range (OHM3000KΩ , 30MΩ ranges are influenced.)

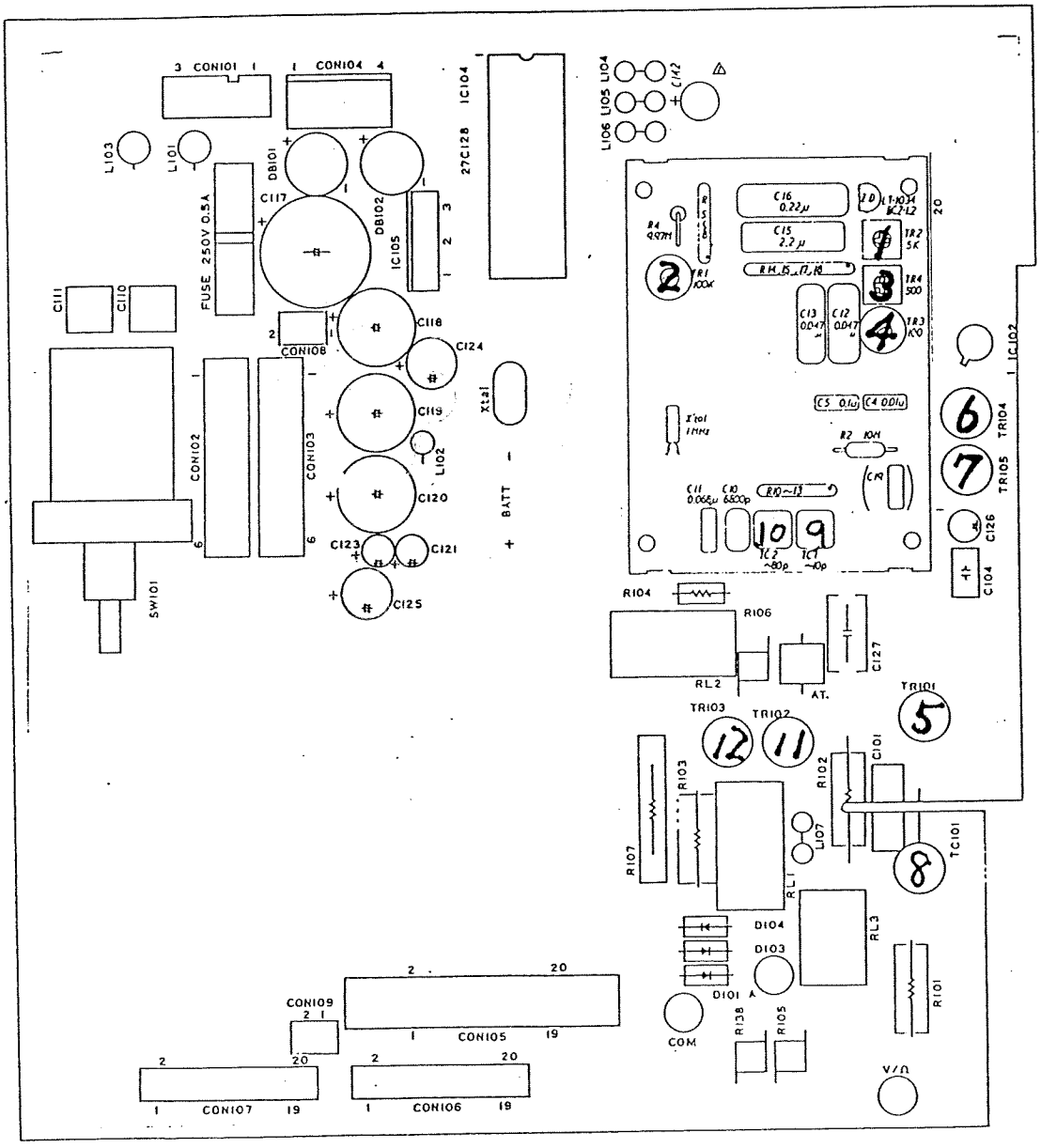
TR102 : To adjust sense point of 300mA shunt resistor (DCA, ACA 300mA ranges are influenced.)

TR103 : To adjust sense point of 3000mA shunt resistor (DCA, ACA 3000mA ranges are influenced.)

TR2 : To adjust integral resistor (all functions/ranges except FRQ function are influenced.)

3）調整手順

No	ﾌｧﾝｸｼｮﾝ,ﾚﾝｼﾞ	入力値	トリマー	調整値
1	OHM,30.000KΩ	29.000KΩ	TR2	29.000KΩ±1dgt
2	OHM,3000.0KΩ	2900.0KΩ	TR1	2900.0KΩ±1dgt
3	DCV,300.00mV	DC 290.00mV COUPLE	TR4 HOSUST.	290.00mV±50dgt ×10
4	DCV,300.00mV	3V DC 290.00mV	TR3	290.00mV±1dgt
5	DCV,1000.0V	DC 1000.0V	TR101	999.8V±1dgt
6	ACA,300.00mA	0	TR104	0.00mA+1dgt
7	ACV,3.0000V	AC 2.9000V 100Hz	TR105	2.9000V±5dgt
8	ACV,300.00V	AC 290.00V 10KHz	TC101	289.80V±5dgt
9	ACV,3.0000V	AC 2.9000V 10KHz	TC1	2.9030V±10dgt
10	ACV,30.000V	AC 29.000V 10KHz	TC2	2.9060V±10dgt
11	DCA,300.00mA	DC 290.00mA	TR102	290.00mA±5dgt
12	DCA,3000.0mA	DC 1000.0mA	TR103	1000.0mA±2dgt



TR4 : To adjust coarse reference voltage (all functions/ ranges except OHM, INΩ, FRQ functions are influenced.)

TR3 : To adjust fine reference voltage (all functions/ ranges except OHM, INΩ, FRQ functions are influenced.)

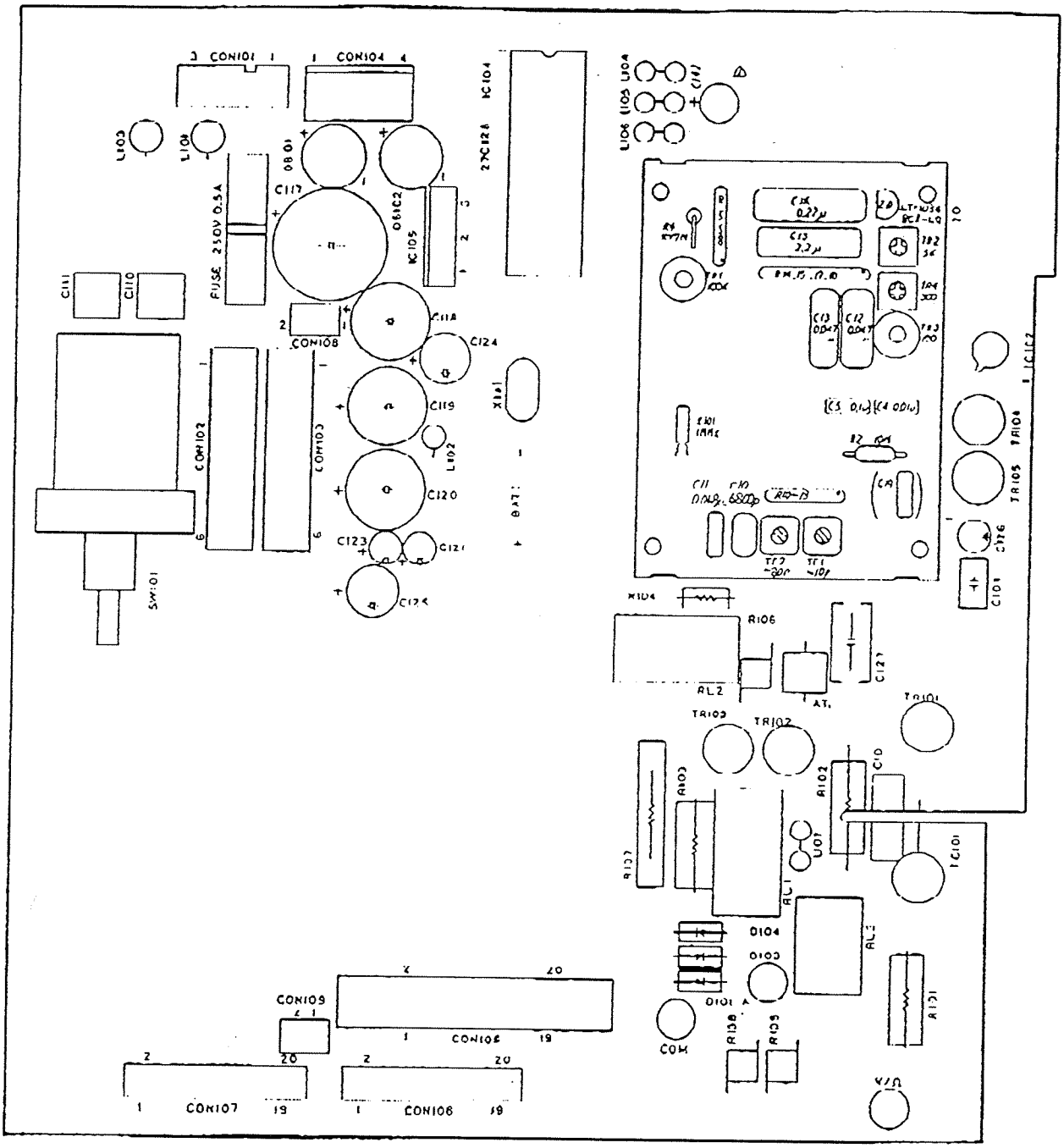
TR104 : To adjust offset of AC/DC converter (all ranges of ACV, ACA are influenced.)

TR105 : To adjust gain of AC/DC converter (all ranges of ACV, ACA are influenced.)

3) ADJUSTMENT PROCEDURES

No	FUNCTION, RANGES	INPUT VALUE	TRIMMER	ADJUSTED VALUE
1	OHM, 30.000KΩ	29.000KΩ	TR2	29.000KΩ ± 1dgt
2	OHM, 3000.0KΩ	2900.0KΩ	TR1	2900.0KΩ + 1dgt
3	DCV, 300.00mV	DC 290.00mV	TR4	290.00mV ± 50dgt
4	DCV, 300.00mV	DC 290.00mV	TR3	290.00mV ± 1dgt
5	DCV, 1000.0V	DC 1000.0V	TR101	999.8V ± 1dgt
6	ACA, 300.00mA	0	TR104	0.00mA + 1dgt
7	ACV, 3.0000V	AC 2.9000V 100Hz	TR105	2.9000V ± 5dgt
8	ACV, 300.00V	AC 290.00V 10KHz	TC101	289.80V ± 5dgt
9	ACV, 3.0000V	AC 2.9000V 10KHz	TC1	2.9030V ± 10dgt
10	ACV, 30.000V	AC 29.000V 10KHz	TC2	2.9060V ± 10dgt
11	DCA, 300.00mA	DC 290.00mA	TR102	290.00mA ± 5dgt
12	DCA, 3000.0mA	DC 1000.0mA	TR103	1000.0mA ± 2dgt

4) PARTS LOCATION



5) TROUBLE SHOOTING

It is recommended that model 856 is repaired on the table treated against electrostatic shock, under grounded through human body, because model 856 has adopted CMOS-LSI in its circuitry.

PHONEMENA	CAUSES
1. POWER LINE/INDICATION	
(1) No Power On (Not Lighted)	·L101~103 ·Line Fuse ·SW101 ·Transformer ·D. B. 102 ·A/D Conv. ·X' tal ·IC101, 104~108, 110~112 ·CON106, 201
(2) Some Indicators not Lighted	·IC201~204 ·LE201~237 ·CON106, 201 ·R201~222 ·Tr201~206
2. FUNCTION	
(1) Function, Range, Function Switch do not work	·IC112, 114 ·R129~136 ·CON107, 202 ·SW201~219 ·D201~219
(2) Somthing Wrong between Communication Equipment and Unit	·IC101, 104, 112, 113 ·A/D Conv. ·Tr104 and its surrondings ·CON105
(3) No Buzzer	·A/D Conv. ·IC201 ·CON106, 201 ·Buzzer
3. DCV	
(1) OFFSET too much	·Leakage on PCB surface ·A/D Conv.
(2) Off Accuracy	·TR2~4, 101 Adjustment Error ·C101 ·TC101 ·R10~13 (5164), 101, 102
(3) No Measurement on all ranges	·R14, 15, 17, 18 (5166), 16, 21, 22 ·ZD ·TR2, 3, 4 ·C4, 5, 12~17 ·A/D Conv.
4. ACV	
(1) OFFSET too much	·Shield, TR104 Adjustment Error ·R110, 111 ·IC102
(2) Off Accuracy	DCV to be checked ·TR105 Adjustment Error
(3) Faulty Frequency Response	·TC101, 1, 2 Adjustment Error ·C8~11, 101 ·IC102
(4) No Measurement on all ranges	·OP1 ·A/D Conv. ·IC102 ·TR104, 105 ·R110, 111 ·C102, 103, 126

PHONEMENA	CAUSES
5. Ω (1) OFFSET too much	·A/D Conv. · Bad Contact in Input Terminal
(2) Off Accuracy (300Ω to 300KΩ range)	·R14, 15 (5160), 5~8 (5165) · TR2 Adjustment Error
(3) Off Accuracy (3000KΩ 30MΩ range)	·TR1 Adjustment Error ·R4 ·Tr101, 102 ·A. T. ·C127
(4) No Measurement on all ranges	·R3, 104~106 ·C12, 13, 127 ·A. T. ·Tr101, 102 ·A/D Conv. ·IC120 ·R121 ·C136, 137
6. DCA (1) OFFSET too much	·R103, 107 ·IC116 ·A/D Conv.
(2) Off Accuracy	·TR102, 103 Adjustment Error ·R103, 107~109 · DC300mV to be checked
(3) No Measurement on all ranges	·Fuse ·D101, 103, 104 ·IC116 ·A/D Conv.
(4) Faulty Range Shift	·R, L1 ·D102 ·IC115, 116 ·Tr103 ·R124, 125
7. ACA	· DCA, ACV & DCA to be checked
8. FRQ (1) Faulty Measurement	·IC103 ·R112, 113, 148 ·C104 ·A/D Conv.
9. DIC (1) Faulty Measurement	·R2 ·A/D Conv. ·DC3V, Ω to be checked
10. INΩ (1) Faulty Measurement	·R19, 20 ·A/D Conv. ·Ω to be checked